

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution																
States both hypotheses using correct notation	AO2.5	B1	H ₀ : Language choice is independent of sex																
Constructs 'correct' contingency table with frequencies shown (PI)	AO3.1b	M1	H ₁ : Language choice is not independent of sex <table><tr><td></td><td>German</td><td>Mandarin</td><td></td></tr><tr><td>M</td><td>7</td><td>12</td><td>19</td></tr><tr><td>F</td><td>23</td><td>8</td><td>31</td></tr><tr><td></td><td>30</td><td>20</td><td>50</td></tr></table>		German	Mandarin		M	7	12	19	F	23	8	31		30	20	50
	German	Mandarin																	
M	7	12	19																
F	23	8	31																
	30	20	50																
Finds expected frequencies (PI)	AO3.4	B1	<table><tr><td></td><td>German</td><td>Mandarin</td></tr><tr><td>M</td><td>11.4</td><td>7.6</td></tr><tr><td>F</td><td>18.6</td><td>12.4</td></tr></table>		German	Mandarin	M	11.4	7.6	F	18.6	12.4							
	German	Mandarin																	
M	11.4	7.6																	
F	18.6	12.4																	
Evaluates $\chi^2 = \sum \frac{(O - E)^2}{E}$ Allow no Yates correction used FT their observed and expected frequencies	AO1.1a	M1F	$\chi^2 = \sum \frac{(O - E - 0.5)^2}{E}$ $= \frac{(3.9)^2}{11.4} + \frac{(3.9)^2}{7.6} + \frac{(3.9)^2}{18.6} + \frac{(3.9)^2}{12.4}$																
Obtains χ^2 test statistic or quotes p -value	AO1.1b	A1	$= 1.33 + 2.00 + 0.82 + 1.23$ ts $\chi^2 = 5.38$ $p = 0.0204$																
Evaluates χ^2 model by comparing test statistic and correct critical value Or by comparing p -value with 0.05	AO3.5a	M1	$df = 1$ $cv = 3.84$ ts > 3.84 or $p < 0.05$																
Infers that H ₀ should be rejected	AO2.2b	E1	Reject H ₀																
Completes test correctly and gives conclusion in context. (conclusion should not be definite)	AO3.2a	E1	Significant evidence to suggest that language choice is not independent of sex																
			Total 8 marks																

Q2.

(a)

O_i	E_i	$(O_i - E_i - 0.5)^2 / E_i$
30	35.2	0.6276
14	8.8	2.5102
130	124.8	0.1770
26	31.2	0.7080
	χ^2	4.0228

E attempted (at least two correct to 1 d.p.)

M1

Yates' correction attempted; at least one correct value in final column

M1

χ^2 attempted

M1

AWFW 4.02 to 4.03

A1

H_0 : No association between method of receiving information and outcome

At least one correct

H_1 : Association between method of receiving information and outcome

If "independent" used, it must be the right way round

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B1

CV of χ^2 for 1 df = 3.84(1)

B1

4.02 > 3.841 so reject H_0

There is significant evidence of an association between method of receiving information and outcome

Dep on A1 and B1 for CV

A1

Applications higher than expected for telephone calls, so council's belief seems to be true

Dep on previous A1

Context conclusion about council's belief, referring to higher than expected for telephone

Alternative if Yates' not used

O_i	E_i	$(O_i - E_i)^2 / E_i$
30	35.2	0.7682
14	8.8	3.0727
130	124.8	0.2167
26	31.2	0.8667
	χ^2	4.9243

Loses M1 for Yates' and A1 for final χ^2 value but can score all the other 6 marks

Final 2 A1 marks dep on 4.92 to 4.93 and B1 for CV

- (b) Type I error was made because

E1

H_0 has been rejected (when it was true)

Dep on previous E1

Edep

SC If ' H_0 accepted' when
their χ^2 less than their CV

No error was made because
 H_0 has been accepted (when it was true)

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(E1)

Dep on previous (E1)

(Edep1)

2

[10]

Q3.

- (a) (i) $E_i: \frac{mp}{N}; \frac{mq}{N}; \frac{np}{N}; \frac{nq}{N}$

B1 any one correct

B2 all correct (simplified)

B2,1

2

- (ii)

$$\begin{aligned}\sum_i E_i &= \frac{mp + mq + np + nq}{N} \\ &= \frac{m(p+q)}{N} + \frac{n(p+q)}{N} \quad (\text{oe}) \\ \sum_i E_i &= \frac{mp + mq + np + nq}{N} \\ &= \frac{m(p+q) + n(p+q)}{N}\end{aligned}$$

(or use of unsimplified forms)

M1

$$\frac{mN}{N} + \frac{nN}{N}$$

$$= m + n$$

$$= \frac{(p+q)(m+n)}{N} = \frac{N \times N}{N} = N$$

Mdep1

$$= N \text{ (since } p + q = m + n = N \text{)} \\ \text{(AG)}$$

Adep1

An example of unsimplified values derived from $a = \frac{mp}{N}$:

$$\Rightarrow b = m - \frac{mp}{N}; c = p - \frac{mp}{N};$$

$$d = n - \frac{mp}{n} \quad (\text{oe})$$

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3

(b) H_0 : No association between Andy's results and wind conditions

B1

E_i :

17.82	15.18	33
9.18	7.82	17
27	23	50

Attempt E's

M1

$$\Rightarrow |0_i - E_i| - 0.5 = 2.32$$

Yates' correction attempted

M1

$$\chi^2 = 0.3020 + 0.3546 + 0.5863 + 0.6883$$

Final column attempted

M1

$$= 1.93$$

awrt

A1

$$\chi^2_{10\%}(1) = 2.706$$

correct value of χ^2 only (allow 2.71)

B1

$\Rightarrow$ Accept H_0

dep (B1 for H_0)

Adep1

No association (between Andy's results and wind conditions)

Appropriate conclusion

dep(B1 for H_0 ; M1final column; $\chi^2_{10\%} = 2.706$.)

Edep1

8

[13]

Q4.

O_i	E_i	$(O_i - E_i - 0.5)$ (α)	$\frac{\alpha^2}{E_i}$
24	28	3.5	0.4375
56	52	3.5	0.2356
11	7	3.5	1.7500
9	13	3.5	0.9423
			3.3654

E attempted

M1

Yates' correction attempted

M1

χ^2 attempted

M1

A1

H_0 : No association between drug and prevention of sickness

H_1 : Association between drug and prevention of sickness

(at least H_0 stated correctly)

B1

$$\chi^2_{5\%} = 3.841$$

CAO

B1

Accept H_0

A1ft

No evidence at the 5% level of significance to support the claim that the drug is effective against sickness.

E1ft

8

[8]

Q5.

(a) H_0 : no association between education and salary

B1

O	E	$\alpha = O - E - 0.5$	$\frac{\alpha^2}{E}$
78	70.2	7.3	0.7591
57	64.8		0.8224
52	59.8		0.8911
63	55.2		0.9654
			3.4380

E attempted, correctly

M1

Yates' correction attempted

A1M1

$\frac{\alpha^2}{E}$ attempted (final col)

M1

AWRT 3.4

A1

$$\chi^2(10\%) = 2.706$$

B1

Reject H_0 at 10% level

A1ft

Evidence to suggest an association
between salary and having a university education.

E1ft

9

(b) Rejecting H_0 when H_0 correct

E1

Stating that there is an association
between salary and education when there is not.

E1

2

[11]

Q6.

(a)

O_i	E_i	$ O_i - E_i - 0.5$	$\frac{7.5^2}{E_i}$
52	44	7.5	1.2784
58	66	7.5	0.8523
28	36	7.5	1.5625
62	54	7.5	1.0417
			4.7349

E attempted

M1

Yates' correction attempted

M1

χ^2 attempted

M1

AWFW 4.73 to 4.74

A1

H_0 : No association between incidence
of asthma and volume of traffic

H_1 : Association

at least H_0 stated correctly

B1

$$v = 1$$

$$\chi^2_{\text{crit}} = 3.841 < 4.7349$$

critical value

B1

Reject H_0 at 5% level

A1ft

Evidence to suggest an association
between the incidence of asthma in
children and the volume of traffic where they live

E1ft

8

- (b) More than expected had asthma
dep on statement of association

E1

1

[9]

Q7.

H_0 : condition independent of treatment
 H_1 : condition dependent upon treatment

B1

Totals: 66, 84, 75, 75

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B1

O	E	$ O - E - 0.5$	$\frac{(O - E - 0.5)^2}{E}$
20	33	12.5	4.7348
55	42		3.7202
46	33		4.7348
29	42		3.7202

for E_i attempted, correctly

M1A1

for use of Yates' correction

M1

final column

M1

$$\chi^2 = 16.91$$

allow 16.9
 If no Yates' correction:
 possible M1A1M0M1A0
 If 0.5 incorrectly used:
 possible M1A1M1M1A0

A1

$$\chi^2_{16.9}(1) = 3.841 < 16.91$$

for χ^2 on their v

B1ft

Reject H_0

iff H_0 stated correctly
 dependent on third M1

A1ft

Evidence to suggest that the condition
 of the patients may be dependent upon
 the treatment that they received

E1ft

10

[10]

Q8.

H_0 : time of day has no effect on the
 outcome of a frame of snooker

H_0 : outcome does not depend on time
 of day

B1

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O_i	E_i	$ O_i - E_i - 0.5$	χ^2
30	25.92	3.58	0.4945
18	22.08	3.58	0.5805
24	28.08	3.58	0.4564
28	23.92	3.58	0.5358
100	100		2.0672

For E's

M1A1

For use of Yates' correction

M1

attempted calculation of χ^2

M1

(even if Yates' correction not used)

A1

$$\chi^2_{5\%}(1) = 3.841$$

For $v = 1$ and χ^2

B1B1ft

$2.07 < 3.841 \therefore$ do not reject H_0

On their χ^2

A1ft

No evidence to suggest that the time of day has an effect on the outcome of a frame of snooker played by Syd.

E1ft

10

[10]

Q9.

H_0 : no association between their gender and type of holiday preferred
(for at least (H_0))

B1

H_1 : there is an association between their gender and type of holiday

Totals:

118	98	216
90	114	204
208	212	420

B1

Expected frequencies:

$$\frac{216 \times 208}{420} = 106.97$$

$$\frac{216 \times 212}{420} = 109.03$$

M1

$$\frac{204 \times 208}{420} = 101.03$$

A1

$$\frac{204 \times 212}{420} = 102.97$$

$$v = 1$$

B1

$$\sum \frac{[O_i - E_i - 0.5]^2}{E_i}$$

Yates' correction

M1

$$= 1.036 + 1.017 + 1.097 + 1.077$$

$$= 4.227:$$

AWFW 4.20 to 4.24

A1

$$\chi^2_{5\%}(1) = 3.841$$

(4.20 if E_i rounded to nearest integer)

B1

$$4.23 > 3.841 \therefore \text{reject } H_0$$

(Only if H_0 stated)

A1ft

Evidence at the 5% level of an association between the gender of single people and the type of holiday that they prefer to take.

E1ft

10

[10]

Q10.

H_0 : no association

OE

B1

120	100	220
24	36	60
144	136	280

correct totals

B1

calculate E_i :

$$\frac{220}{280} \times \frac{144}{280} \times 280 = 113.1$$

$$\frac{220}{280} \times \frac{136}{280} \times 280 = 106.9$$

$$\frac{220}{280} \times \frac{144}{280} \times 280 = 30.9$$

$$\frac{220}{280} \times \frac{144}{280} \times 280 = 29.1$$

Any correct method

M1

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} \quad (\text{Ignore } Y' C)$$

use of

M1

6.357

40.4

0.3572
0.3782

attempted use of Yates' correction

M1

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1.3097
1.3867

*Total = 3.432

$AWFW(3.4 \text{ to } 3.5)$
 $(NO Y' C \Rightarrow 3.993 \Rightarrow A0)$

(*) If wrong answer and no working out then M0 M0 A0

(*) If wrong answer and Yates Correction mentioned then M0
M1

A1

$$\chi^2_{5\%}(1) =$$

CAO ($v = 1$) $\Rightarrow$ by use of 3.841

B1

$$= 3.841$$

ft on their v

B1f.t.

$\therefore$ do not reject H_0
 These results do not suggest
 an association between
 driving schools and test results

Any correct interpretation dependent on 1st M1

A1 f.t.

[9]

Q11.

(a)(b)

	Won	Lost	
< 2 hours	28 22.63	15 20.37	43
≥ 2 hours	22 27.37	30 24.63	52
	50	45	95

Method for table

M1

Labelled correctly, allow >2

A1

Numerically correct

A1

Method for E's – their table

M1

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3

H_0 No association between length of
 match and chance of Boris winning
 H_1 Association between length of match
 and chance of Boris winning

Null hypothesis – may be implied by
 clearly stated conclusion – generous, allow
 1 –sided etc

B1

$$\sum \frac{(|O - E| - 0.5)^2}{E} = 4.04$$

Attempt at $\sum (O - E)^2 / E$

M1

Attempt at Yates' correction

m1

Correct application of Yates' Correction

m1

4.04 (4 ~ 4.10)

A1

c.v. $\chi^2_{1^2}$ is 3.841

1 df

B1ft

3.841 or 3.84, their df

B1ft

Reject H_0 , evidence of association
between result and length of game.

*ft their figures – needs all M Marks and
must be compared with upper tail of χ^2
(maximum 8 out of 9 if method of
constructing table is incorrect) – Needs
clearly stated correct conclusion or
hypothesis*

A1ft

9

- (c) There is evidence of association between
result and length of game but contrary to
Boris's belief he appears to be less likely
to win longer games.

evidence of association

E1

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*Boris less likely to win long games (2
marks for this point if evidence of
association clearly stated in (b))
Needs proportion probably implied for
both marks*

E1

2

[14]

Q12.

(a)

	D	P	DN
< 65	4 2.70	12 7.72	6 11.58
≥ 65	10 11.30	28 32.28	54 48.42

Method for E's

M1

Pooling D&P

	D&P		DN	
< 65	16	10.42	6	11.58
≥ 65	38	43.58	54	48.42

correct method for pooling - must be D&P

m1

H_0 : Answer not associated with age

B1

H_1 : Answer associated with age

correct null hypothesis - may be implied by clearly stated conclusion

$$\frac{(|O-E| - 0.5)^2}{E} =$$

attempt at $\sum \frac{(O-E)^2}{E}$ - their figures

M1

attempt at Yates' correction

m1

correct method for Yates'

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m1

$$\frac{5.08^2}{\left(\frac{1}{10.42} + \frac{1}{11.58} + \frac{1}{43.58} + \frac{1}{48.42} \right)} = 5.83$$

5.83 (5.81 to 5.85)

A1

c.v χ^2_1 is 3.841

1df or 2df if no pooling

B1

3.841 or 3.84 (allow 5.991/5.99 if no pooling)

B1

Reject H_0 , conclude evidence to suggest

that answer is associated with age.

correct conclusion - must be compared with upper tail of χ^2

A1ft

10

- (b) Older delegates more likely to answer that they would not change their votes.

Older delegates less likely to answer that they would change their votes

E1

1

- (c) Most delegates are over 65

Most delegates are elderly

E1

1

[12]

Q13.

- (a) H_0 : Reason not associated with time of day

Correct null hypothesis – may be implied by clearly stated conclusion

B1

H_1 : Reason associated with time of day

	Rush hour		Out of RH	
Normal	34	27.05	45	51.95

Method for one E

M1

Beatles	4	10.95	28	21.05
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Method for all E's

m1

$$\frac{\sum \frac{(|O - E| - 0.5)^2}{E}}{\text{Attempt}} = \frac{\sum (O - E)^2}{E}$$

Attempt Yates's correction

Clearly correct method for Yates's

M1 m1 m1

$$6.45^2 \left(\frac{1}{27.05} + \frac{1}{51.95} + \frac{1}{10.95} + \frac{1}{21.05} \right) = 8.11$$

$$8.11 (8.05 - 8.15)$$

A1

c.v. χ^2 is 3.841

1 d.f.

3.841 or 3.84

B1 B1ft

Reject H_0 conclude reason associated with time of day

Conclusion – t.s. must be compared with upper tail of χ^2 - needs first 3 M marks

A1ft

10

- (b) Of those using the crossing because of the Beatles, a greater proportion do so out of rush hour than during rush hour

Greater **proportion** out of rush hour

E1

1

[11]

Q14.

- (a) (i) H_0 : Irritation not associated with carrying phone

Correct null hypothesis - if not stated may be implied by final correct conclusion

 H_1 : Irritation associated with carrying phone

Needs clear statement; can be earned in conclusion or (a)(ii)

	Irritated		Not irritated	
Phone	12	17.03	14	8.97
No phone	26	20.97	6	11.03

Correct method for one E

B1

Correct method for all E's

M1 M1

$$\sum \frac{(|O - E| - 0.5)^2}{E}$$

$$\sum \frac{(O - E)^2}{E}$$

Attempt

Attempt at Yates's correction

M1 m1

$$= 4.53^2 \left(\frac{1}{17.03} + \frac{1}{8.97} + \frac{1}{20.97} + \frac{1}{11.03} \right) = 6.33$$

Correct method for Yates's

(6.2 - 6.5) or demonstration

$$\sum \frac{(O - E)^2}{E}$$

exceeds their c.v.

m1 A1

$$cv \chi^2_{1} 5\% = 3.841$$

for 1 degree of freedom

3.841 or 3.84

B1 B1

Reject H_0 : irritation associated with carrying mobile phone.

Correct conclusion – requires first M1m1M1 and

upper tail of χ^2 used

A1ft

10

- (ii) Passengers not carrying mobile phones more likely to be irritated than those carrying mobile phones

Conclusion in context

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E1

1

- (b) (i) Combining dissimilar classes makes interpretation difficult or impossible

Interpretation difficult

E1

1

- (ii) Not necessary to combine extremely irritated as $E > 5$

Not necessary $E > 5$

E1

Anthony's suggestion leaves 3×2 table to be analysed which is more likely to establish an association if one exists (more degrees of freedom).

More likely to establish a difference / more degrees of freedom

(iii) H_0 : Irritation not associated with carrying phone

H_1 : Irritation associated with carrying phone

Null hypothesis - if not stated may be implied by final correct conclusion

B1

	Not / Slightly		Irritated		Extremely	
Phone	52	39.70	18	20.49	3	12.81
No Phone	10	22.30	14	11.51	17	7.19

$$\sum \frac{(O - E)^2}{E}$$

Attempt at $\sum \frac{(O - E)^2}{E}$

M1 M1

$$= \frac{12.30^2}{39.70} + \frac{12.30^2}{22.30} + \frac{2.49^2}{20.49} + \frac{2.49^2}{11.51} + \frac{9.81^2}{12.81} + \frac{9.81^2}{7.19} = 32.3$$

(32.2 - 32.5) or demonstration

$$\sum \frac{(O - E)^2}{E} \text{ exceeds their cv}$$

A1

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$\chi^2_{2, 1\%} = 9.210$
 2 degrees of freedom
 9.210 or 9.21

B1 B1

Reject H_0 , irritation associated with carrying phone.

Correct conclusion - requires previous M1 only and upper tail of χ^2 used.

A1ft

Passengers not carrying mobile phones more irritated than others

Conclusion in context

E1

Examiner reports

Q2.

Although the calculation of the test statistic was generally well done, with most students applying Yates' correction, very few truly appreciated what the χ^2 -test showed. The hypotheses were often in terms of the phone call, and even where the hypotheses were in terms of 'method of contact' the final conclusion was often "showing that a phone call is more effective". Only the best students realised that the χ^2 -test could only establish that an association between 'method of contact' and 'application for grant' existed, and that the values in the table must then be examined to support the council's belief. The majority correctly identified 'Type I error' and the reason for it.

Q3.

Part (a) was designed to test candidates' understanding of the properties of a contingency table. This proved to be very popular, with the majority of candidates showing a good understanding and demonstrating good algebraic skills to often achieve full marks. In part (b), most realised that Yates' correction was required even if they could not apply it correctly in all cases. However, this did, as usual, prove to be a popular source of marks, even for those candidates who found part (a) beyond them.

Q4.

This was the most common source of marks for the majority of candidates, with many fully-correct solutions seen. However, too many candidates still failed to state their hypotheses but nevertheless felt justified in drawing some sort of conclusion. Since a 2×2 table was given, the use of Yates' correction was required, something that some candidates failed to do.

Q5.

This, once again, proved to be a very popular topic and an excellent source of marks. Although there were still a few candidates who failed to state any hypotheses at all, those who did usually did so correctly. Most realised that $v = 1$ and that Yates' correction was required. Unfortunately, there were some who could not apply this correction in the correct way. Although the majority of candidates correctly considered $|O_i - E_i| - 0.5 = 7.3$, and

then went on to calculate $X^2 = \sum_i \left(\frac{7.3^2}{E_i} \right)$, there were some who thought wrongly that $X^2 = \sum_i \left\{ \frac{(O_i - E_i - 0.5)^2}{E_i} \right\}$ or even that $X^2 = \sum_i \left\{ \frac{((O_i - E_i)^2 - 0.5)}{E_i} \right\}$ were appropriate.

It was pleasing to see good conclusions in context.

In part (b), most candidates understood what a Type I error was, stating that this occurred when " H_0 was rejected when in fact H_0 was correct". Unfortunately, many could not then interpret this Type I error in the context of the question.

Q6.

Although this first question provided most candidates with a good start to the paper, there were some who, in part (a), either failed to state any hypotheses or who thought that the

null hypothesis should indicate ‘an association’, rather than ‘no association’, between the incidence of asthma and the volume of traffic. Also simply stating “ H_0 : not associated” and “ H_1 : associated” was not really sufficient when a statement in the context of the question was required. Although the great majority of candidates realised that Yates’ correction had to be used, there were far too many candidates who could not apply it correctly. A conclusion in context was required: simply stating “Reject H_0 ” was not sufficient to gain full marks. Although most candidates formulated a correct conclusion in context, some either used statements which were too positive in nature or made no statement at all. In part (b), the question asked for a comment about the 52 children who had asthma, given that they lived in an area where the volume of traffic was heavy. It did **not** ask, as many candidates thought, for a comparison with the children who lived in an area where the volume of traffic was light. Consequently, the required response “More than expected had asthma” was not always seen. Some candidates, having drawn the wrong conclusion in part (a) that there was ‘no association’ between the incidence of asthma and the volume of traffic, still felt justified in making this statement.

Q7.

This question, as usual on this topic, proved to be a good source of marks for the majority of candidates. However, there were still some candidates who either did not state their hypotheses in context or omitted them completely. Simply stating “ H_0 : not associated” and “ H_1 : associated” was not sufficient. Although the vast majority of candidates realised that Yates’s correction had to be used, there was a great number of candidates who could not apply it correctly. The most common incorrect applications were $O_i - E_i - 0.5$ or $(O_i - E_i)^2 - 0.5$, whereas $|O_i - E_i| - 0.5$ was required. Although most candidates formulated a conclusion in context, these were usually too positive in nature. Also, simply stating “Reject H_0 ” was not sufficient to gain full marks.

Q8.

This was the best answered question on the paper, with very many fully correct solutions seen. However, some candidates did not state their hypotheses but still felt justified in making statements such as ‘do not reject H_0 ’ as a conclusion. Although most candidates realised that Yates’ correction was required, a few failed to apply it correctly. Some of the candidates, $|O_i - E_i - 0.5|$ was used in the evaluation of χ^2 instead of $|O_i - E_i|$. A minority of candidates looked values up incorrectly from Table 6 or even used the wrong tables. A conclusion in context, which is always required in questions of this type, was not always evident.

Q9.

Solutions to this question were, as usual, mostly excellent. However, some candidates still failed to state their hypotheses whilst others got H_0 and H_1 the wrong way round. Candidates should realise that stating ‘reject H_0 ’ without first stating H_0 is inappropriate. Also expected values should not be rounded to whole numbers especially when the value of χ^2 is then quoted to two or more places of decimals. Few candidates failed to attempt to

use Yates’ correction but several candidates considered $\frac{(O_i - E_i - 0.5)^2}{E_i^1}$ rather than the

correct form of $\frac{(|O_i - E_i| - 0.5)^2}{E_i}$ when applying the correction. A statement of a conclusion in context was expected but was not always seen.

Q10.

Candidates scored consistently well on this question, with many gaining full marks. However, some still failed to state their hypotheses while others wrongly assumed that H_0 indicated an association between test results and the school chosen.

Only a few candidates omitted the use of Yates' correction whilst even fewer did not state $v = 1$, leading to a χ^2 value of 3.841.

Candidates should be encouraged to write their conclusions in the context of the problem set. Statements such as 'do not reject H_0 ' or 'accept H_1 ', without reference to the context of the question are not likely to gain full credit.

Q11.

As usual this type of question was extremely well answered with the great majority of candidates constructing the contingency table correctly and most applying Yates' correction accurately. The most common error with Yates' correction was to use $(O - E - 0.5)$ instead of $(|O - E| - 0.5)$. There were some problems with the null hypothesis. Apart from some nonsensical hypotheses such as 'no difference between the length of the game and the probability of winning' there were also some one-sided hypotheses such as 'the longer a match lasts the better the chance of Boris winning'. The hypothesis test can only test for 'association' although further interpretation is possible by examining the table.

Q12.

As usual candidates were well able to deal with contingency tables although some lost marks by failing to pool. A few pooled without first calculating any expected values, implying that they had based their decision to pool on the observed rather than the expected values.

Q13.

Nearly all candidates were able to make a good attempt at analysing the contingency table. Only a few failed to attempt Yates' correction, although the use of

$\sum \frac{(O - E - 0.5)^2}{E}$ instead of $\sum \frac{(|O - E| - 0.5)^2}{E}$ was not uncommon. However, the interpretation in part (b) was not well done. The expected answer was that the proportion of people using the crossing because of the Beatles was less in rush hour than outside rush hour.

Q14.

Most candidates scored highly on this question. The mechanics of analysing contingency tables, including, where appropriate, combining classes and the use of Yates's correction, seems well understood. Many had difficulty framing appropriate hypotheses. ' H_0 not linked: H_1 linked' is not adequate, although candidates were allowed to redeem themselves by a clearly stated conclusion which enlarged on exactly what was linked. In part (a)(ii), candidates were expected not only to state that irritation and carrying a mobile phone were associated but that (unsurprisingly) passengers not carrying a mobile phone were more likely to be irritated than those carrying a mobile phone.

In parts (b)(i) and (b)(ii), many candidates scored marks by having more or less the right ideas despite some difficulty in expressing them convincingly. In part (b)(i), most candidates thought it was a bad idea to combine opposites while not going on to comment on the difficulty this would cause when trying to interpret the results. In part (b)(ii), most candidates thought a 3×2 table was preferable to a 2×2 table, without really being sure

why. Vague reasons such as 'more accurate' were given. More substantial reasons such as 'less likely to lead to a Type 2 error' were much rarer.



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